

AI-Driven Cost Predictor in the Non-Life Insurance Sector

HarvardX Data Science Professional Certificate PH125.9x Capstone

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Contents

Introduction	3
Understanding Car Insurance	3
How Insurance Works	3
Impact of the Digital Age	3
The Role of Artificial Intelligence	3
Objective	4
Claims Prediction	4
Cost Prediction	4
Customer Fidelity	4
Data Limitations	4
Dataset	5
The Rarity of Open Datasets in the Insurance Sector	5
Selected Dataset	5
Structure	5
Exploratory Data Analysis	10
Policies	10
Correlations	11
References	23

Introduction

The insurance industry serves as a critical safety net in today's society, offering a mechanism for individuals and businesses to manage risk. At its core, insurance is a contractual agreement between an insurer and a policyholder, where the insurer provides financial protection against certain risks in exchange for regular payments known as premiums. This mitigates the financial impact of unforeseen events, such as accidents, natural disasters, or health issues, ensuring that policyholders are not left vulnerable in times of need.

Understanding Car Insurance

Car insurance specifically addresses risks related to vehicle ownership and operation. It provides coverage for damages to the vehicle, bodily injury to the driver or passengers, and third-party liabilities that may arise from motor vehicle accidents. Policies can vary widely, including comprehensive coverage, which protects against theft or damage outside of collisions, and liability coverage that pays for damages to other parties in the event of an accident caused by the policyholder.

How Insurance Works

The fundamental principle of insurance is risk pooling. By collecting premiums from multiple policyholders, insurance companies can create a reserve fund to cover claims when they arise. The likelihood of individual claims is statistically evaluated, allowing insurers to set premium rates that reflect the risk associated with different policyholders. This system is crucial for maintaining financial stability within the insurance sector, enabling insurers to meet their commitments even in challenging circumstances.

Impact of the Digital Age

In recent years, the insurance industry has experienced significant transformation due to the digital age. Technologies such as mobile applications, online platforms, and telematics have revolutionized how insurance is marketed, sold, and managed. Customers now enjoy convenience in purchasing policies, accessing services, and making claims through user-friendly digital interfaces.

The Role of Artificial Intelligence

Among the most influential advancements in the insurance sector is the integration of artificial intelligence (AI) [1]. AI-enhanced systems can analyze vast amounts of data in real-time, improving underwriting processes, claim assessments, and customer service interactions. In car insurance, AI algorithms can evaluate driving behavior through telematics devices, tailoring coverage and pricing based on actual risk rather than historical averages. This not only leads to personalized premium rates but also encourages safer driving habits among policyholders.

Furthermore, AI enables predictive analytics, which empowers insurance companies to anticipate future risks and trends. By leveraging data from various sources, including social media, weather patterns, and historical claims, insurers can develop cost prediction models that enhance financial forecasting and optimize reserve allocation. This transformative approach helps insurance companies remain competitive while also improving customer satisfaction through tailored services.

Objective

This document represents my final capstone project for the HarvardX Data Science Professional Certificate. Throughout this course, I have acquired valuable skills in data analysis, statistical modeling, and machine learning [2]. This project aims to apply those skills within the insurance industry by examining various predictive aspects related to car insurance.

The primary focus of this report is to utilize machine learning tools to predict several key aspects of car insurance, specifically the costs associated with insurance policies, the number of claims, and customer loyalty. By integrating theoretical knowledge with practical applications, I seek to demonstrate my proficiency and understanding gained during the course. Harnessing advanced algorithms and data analytics, this project aspires to deliver insightful predictions that account for multiple risk factors impacting policyholders and their individual situations.

Claims Prediction

Understanding the number of claims a policyholder is likely to make is crucial for insurers. The frequency and severity of claims directly influence the overall cost of insurance, as a history of frequent claims generally categorizes an individual as a higher risk. By predicting the number of claims accurately, insurers can better manage their risk profiles and set more precise premiums.

Cost Prediction

The cost of insurance can vary significantly based on several factors, including the type of coverage selected and the individual's claims history. For example, comprehensive coverage, which protects against various risks beyond mere collisions, typically incurs higher premiums compared to basic liability coverage. This section will utilize machine learning algorithms to forecast insurance costs by analyzing different contributing factors aligned with policyholders' unique risk profiles.

Customer Fidelity

Customer fidelity plays a pivotal role in influencing insurance costs. Research indicates that policyholders who maintain their insurance over extended periods are often rewarded with lower premiums. Insurers perceive long-term relationships as indicative of lower risk based on more stable claims histories and overall customer loyalty. This section will explore metrics that assess customer fidelity and its impact on insurance pricing.

Data Limitations

Despite the ambition to create a robust predictive model, this project faces notable data limitations. The dataset contains limited demographic and policy-related information, and the timeframe includes only 4 to 5 years of data. These constraints can hinder accurate assessments of early termination risk and comprehensive customer profiling. Nevertheless, the goal remains to maximize the use of available data, focusing on the most relevant features to offer meaningful insights into insurance costs and risk assessment in subsequent applications.

Dataset

The Rarity of Open Datasets in the Insurance Sector

Open datasets in the insurance industry are relatively scarce due to the highly sensitive nature of the information they contain. Insurance companies manage vast amounts of personal and financial data, which, if exposed, could compromise individual privacy and lead to significant legal consequences. The insurance sector is also heavily regulated, requiring strict compliance with data protection laws, such as GDPR in Europe. This regulatory framework makes insurers particularly cautious about sharing their data publicly.

Furthermore, insurance data is often perceived as a competitive advantage. Companies are reluctant to disclose their datasets, fearing that it could reveal proprietary algorithms and models designed for specific underwriting processes and claims assessments. As a result, high-quality open datasets that could significantly contribute to academic and research endeavors are limited.

Selected Dataset

Given these challenges, the dataset I have selected for analysis is exceptional due to its comprehensive nature and applicability in evaluating AI-driven cost prediction models in the car insurance sector. This dataset originates from a **Spanish insurance company that specializes in non-life insurance**.

While the dataset does include some policyholder demographic information, it is notably limited in scope. However, it features a variety of other important variables essential for a thorough analysis of risks, premiums, and factors influencing claims. This includes details related to the vehicles insured, historical claims data, and additional criteria that can impact risk assessment.

Accessing such a dataset is valuable as it allows for the exploration of correlations and trends relevant to AI-driven cost predictors. The dataset is publicly available at the Mendeley Data website [3], but to ensure its continued accessibility, I have also copied the dataset to my GitHub repository [4]. This redundancy will ensure that the dataset remains available even if the original website removes it. This data is an important resource for understanding the complexities of car insurance and enables the application of predictive analytics in this domain. Through this analysis, I aim to uncover insights that will enhance the development and implementation of AI-driven cost models in insurance.

Structure

The dataset is organized as three different files : a spreadsheet and two comma-separated value files.

 Descriptive of the variables.xlsx

 Motor vehicle insurance data.csv

 sample type claim.csv

The main dataset is in the “Motor vehicle insurance data.csv” file, comprising **105,555 observations** across **30 variables**. Each row within the dataset corresponds to an individual insurance policy recorded over a specific year, covering the period from **November 2015 to December 2018**.

observations	variables
105,555	30

The table below is based on the “Descriptive of the variables” spreadsheet, indicating the different variables present in the dataset.

Variables	Description
ID	Internal identification number assigned to each annual contract formalized by an insured. Each policyholder can have multiple rows in the dataset, representing different annuities of the product.
Date_start_contract	Start date of the policyholder’s contract (DD/MM/YYYY).
Date_last_renewal	Date of last contract renewal (DD/MM/YYYY).
Date_next_renewal	Date of the next contract renewal (DD/MM/YYYY).
Distribution_channel	Classifies the channel through which the policy was contracted. 0 for Agent and 1 for Insurance brokers.
Date_birth	Date of birth of the insured declared in the policy (DD/MM/YYYY).
Date_driving_licence	Date of issuance of the insured person’s driver’s license (DD/MM/YYYY).
Seniority	Total number of years that the insured has been associated with the insurance entity, indicating their level of seniority.
Policies_in_force	Total number of policies held by the insured in the insurance entity during the reference period.
Max_policies	Maximum number of policies that the insured has ever had in force with the insurance entity.
Max_products	Maximum number of products that the insured has simultaneously held at any given point in time.
Lapse	Number of policies that the customer has cancelled or has been cancelled for nonpayment in the current year of maturity, excluding those that have been replaced by another policy.
Date_lapse	Lapse date. Date of contract termination (DD/MM/YYYY).
Payment	Last payment method of the reference policy. 1 represents a half-yearly administrative process and 0 indicates an annual payment method.
Premium	Net premium amount associated with the policy during the current year.
Cost_claims_year	Total cost of claims for the insurance policy during the current year.
N_claims_year	Total number of claims incurred for the insurance policy during the current year.
N_claims_history	Total number of claims filed throughout the entire duration of the insurance policy.
R_Claims_history	Ratio of the number of claims filed for the specific policy to the total duration (whole years) of the policy in force. It provides an indication of the policy’s claims frequency history.
Type_risk	Type of risk associated with the policy. Each value corresponds to a specific risk type: 1 for motorbikes, 2 for vans, 3 for passenger cars and 4 for agricultural vehicles
Area	Dichotomous variable indicates the area. 0 for rural and 1 for urban (more than 30,000 inhabitants) in terms of traffic conditions.
Second_driver	1 if there are multiple regular drivers declared, or 0 if only one driver is declared.
Year_matriculation	Year of registration of the vehicle (YYYY).
Power	Vehicle power measured in horsepower.
Cylinder_capacity	Cylinder capacity of the vehicle.
Value_vehicle	Market value of the vehicle on 31/12/2019.
N_doors	Number of vehicle doors.
Type_fuel	Specific kind of energy source used to power a vehicle. Petrol (P) or Diesel (D).
Length	Length, in meters, of the vehicle.
Weight	Weight, in kilograms, of the vehicle.

This table indicates that we have very limited information about the policyholder (or driver). Specifically, we only possess details regarding their age, the duration of their driving licence, the type of car they are driving, and their geographic area. However, we lack critical information such as gender, occupation, and income. In contrast, we have more comprehensive details about the car itself, including its power, number of doors, length, and weight, but we do not have the make or model. Additionally, we have some information related to the insurance policy.

Here is a summary of the variables associated with the drivers :

Date_birth	Date_driving_licence	Seniority	Area	Second_driver
Min. :1918-04-05	Min. :1942-10-01	Min. : 1.000	Min. :0.0000	Min. :0.0000
1st Qu.:1960-11-10	1st Qu.:1983-04-13	1st Qu.: 3.000	1st Qu.:0.0000	1st Qu.:0.0000
Median :1970-08-11	Median :1993-07-16	Median : 4.000	Median :0.0000	Median :0.0000
Mean :1970-02-20	Mean :1992-09-12	Mean : 6.697	Mean :0.2739	Mean :0.1237
3rd Qu.:1980-03-03	3rd Qu.:2003-01-24	3rd Qu.: 9.000	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :2000-10-11	Max. :2018-11-26	Max. :40.000	Max. :1.0000	Max. :1.0000

The date of birth ranges from April 5, 1918 to October 11, 2000, indicating that the drivers cover a wide span of ages, with most being born between 1960 and 1980. This suggests that the dataset includes both older and relatively newer drivers, allowing for an analysis of trends across different age groups.

The date of driving license acquisition varies from October 1, 1942 to November 26, 2018, indicating that some drivers have held their licenses for many decades, while others are more recent license holders.

Seniority ranges from 1 to 40 years, with an average of about 6.7 years. This suggests a mix of novice and experienced policyholders, although the median seniority of 4 years indicates that many drivers are relatively new to their insurance policies.

The area variable is binary, indicating whether the drivers are located in urban (1) or rural (0) settings. Approximately 27.39% of drivers reside in urban areas, highlighting a predominance of rural drivers in the dataset.

Lastly, the second driver variable shows that only about 12.37% of the drivers have a secondary driver listed, indicating that most operate their vehicle independently.

We have some additional variables related to the vehicles :

N_doors	Power	Cylinder_capacity	Type_fuel	Length	Weight
Min. :0.000	Min. : 0.00	Min. : 49	D :64998	Min. :1.978	Min. : 43
1st Qu.:3.000	1st Qu.: 75.00	1st Qu.:1390	P :38793	1st Qu.:3.999	1st Qu.:1043
Median :5.000	Median : 90.00	Median :1598	NA's: 1764	Median :4.230	Median :1205
Mean :4.068	Mean : 92.68	Mean :1618	NA	Mean :4.252	Mean :1191
3rd Qu.:5.000	3rd Qu.:110.00	3rd Qu.:1910	NA	3rd Qu.:4.443	3rd Qu.:1388
Max. :6.000	Max. :580.00	Max. :7480	NA	Max. :8.218	Max. :7300
NA	NA	NA	NA	NA's :10329	NA

Year_matriculation	Value_vehicle
Min. :1950	Min. : 270.5
1st Qu.:2001	1st Qu.: 13127.2
Median :2005	Median : 17608.8
Mean :2005	Mean : 18413.7
3rd Qu.:2008	3rd Qu.: 22595.0
Max. :2018	Max. :220675.8

The number of doors on the cars ranges from 0 to 6, with a median of 5 doors and an average of approximately 4.07, suggesting that most vehicles are likely standard sedans or hatchbacks. The power output varies significantly, from a minimum of 0 to a maximum of 580 horsepower, with a mean of 92.68. This indicates a diverse range of vehicle types, from compact cars to high-performance models. The cylinder capacity ranges from 49 to 7480 cubic centimeters, with an average of 1618 cm³, reflecting varying engine sizes suited to different performance needs and fuel types.

When examining fuel types, a significant majority of cars in the dataset use diesel (D), with 64,998 instances, while 38,793 vehicles are powered by petrol (P), and 1,764 entries have missing data for fuel type. The length

of the vehicles spans from 1.978 to 8.218 meters, with a mean length of approximately 4.25, indicating a prevalence of standard-sized cars. The weight ranges from 43 to 7300 kilograms, with a mean of 1191 kg, suggesting that the dataset includes both lightweight cars and heavier vehicles, potentially such as vans or trucks.

In terms of year of matriculation, the cars range from models dated 1950 to 2018, with a mean and median year around 2005, indicating that the dataset covers both classic and more modern vehicles. Finally, there is a wide variance in the value of the vehicles, with a minimum of approximately 270.5 and a maximum of around 220,675.8. The average vehicle value is approximately 18,413.7, reflecting a mixture of economical and luxury cars in the collection.

Now the policy details :

ID	Date_start_contract	Date_last_renewal	Date_next_renewal	Distribution_channel
Min. : 1	Min. :1980-10-25	Min. :2015-11-02	Min. :2016-11-02	Min. :0.0000
1st Qu.:12925	1st Qu.:2013-04-19	1st Qu.:2016-09-05	1st Qu.:2017-09-05	1st Qu.:0.0000
Median :26082	Median :2015-09-20	Median :2017-07-06	Median :2018-07-06	Median :0.0000
Mean :26271	Mean :2014-06-26	Mean :2017-06-19	Mean :2018-06-19	Mean :0.4513
3rd Qu.:39754	3rd Qu.:2017-03-01	3rd Qu.:2018-04-06	3rd Qu.:2019-04-06	3rd Qu.:1.0000
Max. :53502	Max. :2018-11-30	Max. :2018-11-30	Max. :2019-11-30	Max. :1.0000

Policies_in_force	Max_policies	Max_products	Lapse	Date_lapse
Min. : 1.000	Min. : 1.000	Min. :1.000	Min. :0.0000	Min. :2015-11-02
1st Qu.: 1.000	1st Qu.: 1.000	1st Qu.:1.000	1st Qu.:0.0000	1st Qu.:2017-02-03
Median : 1.000	Median : 2.000	Median :1.000	Median :0.0000	Median :2017-10-02
Mean : 1.456	Mean : 1.837	Mean :1.066	Mean :0.2218	Mean :2017-09-07
3rd Qu.: 2.000	3rd Qu.: 2.000	3rd Qu.:1.000	3rd Qu.:0.0000	3rd Qu.:2018-05-19
Max. :17.000	Max. :17.000	Max. :4.000	Max. :7.0000	Max. :2019-06-01
NA	NA	NA	NA	NA's :70408

Payment	Premium	Type_risk
Min. :0.0000	Min. : 40.14	Min. :1.000
1st Qu.:0.0000	1st Qu.: 241.61	1st Qu.:3.000
Median :0.0000	Median : 292.28	Median :3.000
Mean :0.3192	Mean : 315.89	Mean :2.722
3rd Qu.:1.0000	3rd Qu.: 361.64	3rd Qu.:3.000
Max. :1.0000	Max. :2993.34	Max. :4.000

Cost_claims_year	N_claims_year	N_claims_history	R_Claims_history
Min. : 0.0	Min. : 0.0000	Min. : 0.000	Min. : 0.0000
1st Qu.: 0.0	1st Qu.: 0.0000	1st Qu.: 0.000	1st Qu.: 0.0000
Median : 0.0	Median : 0.0000	Median : 1.000	Median : 0.0800
Mean : 153.6	Mean : 0.3947	Mean : 2.752	Mean : 0.4252
3rd Qu.: 0.0	3rd Qu.: 0.0000	3rd Qu.: 4.000	3rd Qu.: 0.5900
Max. :260853.2	Max. :25.0000	Max. :52.000	Max. :26.0700

The policy IDs range from 1 to 53,502, indicating a substantial dataset of individual insurance policies. The contract start dates span from October 25, 1980, to November 30, 2018, with a mean date around June 26, 2014, suggesting a blend of older and more recent policies. The last renewal dates have a similar range, with the earliest being November 2, 2015, and the latest also being November 30, 2018, implying regular renewals among many policies. The next renewal dates range from November 2, 2016, to November 30, 2019, with a mean of June 19, 2018, highlighting trends in customer retention and policy renewals.

Most policyholders maintain 1 to 2 policies, although some hold up to 17 policies, indicating that while many keep a single insurance policy, there are individuals with multiple coverages. The number of claims per policy varies, with a maximum of 25 claims per year and an average of about 0.39 claims, suggesting that most policyholders do not frequently file claims. The cost of claims ranges from 0 to approximately 260,853.2, with an average claim cost around 153.6, indicating that many claims are not high-cost but there are outliers with significant expenses.

Premium costs span from approximately 40.14 to 2,993.34, with a mean premium of about 315.89, reflecting the variety of coverage levels available. The type of risk associated with the policies corresponds to specific vehicle categories: 1 for motorbikes, 2 for vans, 3 for passenger cars, and 4 for agricultural vehicles, with a predominance of risk type 3 (passenger cars) in the dataset.

The last file “sample type claim.csv” contains a list of different claims. It has 7,366 observations or claims while the main dataset contains 19,646 observations with at least one claim.

As its name suggests, it is a sample and won’t be useful for the rest of the analysis.

Exploratory Data Analysis

Policies

Number of different policies per year and total :

Year	Days	Policies	Policies with claims	Total claims	Total Cost
2015	59	4,559	1,367	3,240	1,192,965
2016	365	31,428	8,888	20,862	7,749,565
2017	364	33,753	6,004	12,404	4,947,976
2018	333	35,815	3,387	5,156	2,318,236
Total	1,124	53,502	15,675	41,662	16,208,741

In 2015, we recorded only **4,559** policies, as the data starts from November of that year. In **2016**, the number of unique policies rose to **31,428**. This growth continued in **2017**, reaching **33,753** unique policies, and further increased in **2018** to a total of **35,815** unique IDs. Overall, the total number of unique policies renewed across these four years amounts to **53,502**.

Year	Total Policies	New Policies	Terminating Policies	Proportion All With Claims	Proportion New With Claims	Proportion Terminating With Claims
2015	4,559	1,112	756	0.300	0.417	0.003
2016	31,428	7,709	5,674	0.283	0.406	0.044
2017	33,753	9,505	5,636	0.178	0.346	0.030
2018	35,815	11,343	4,654	0.095	0.182	0.033

The data provides a comprehensive overview of policy trends over a four-year period from 2015 to 2018, detailing total policies, new and terminating policies, and their associated claims proportions. We can see that most of the data come from pre existing policies (created years before 2015) probably loaded later in the dataset.

In 2015, there were 4,559 total policies, with 1,112 new policies introduced and 756 terminating policies. The proportion of all policies with claims was about 30%, suggesting a significant level of risk or incidents among these policies. Notably, 41.7% of new policies in 2015 filed claims, indicating a relatively high incidence of claims within newly established contracts. In contrast, only 0.3% of terminating policies had claims, highlighting a considerably lower risk level as these contracts concluded.

As the years progressed, total policies saw a sharp increase, peaking at 35,815 in 2018. The proportion of policies with claims, however, revealed a downward trend, falling to 9.5% by 2018. This decline in overall claims might suggest improvements in risk management, underwriting criteria, or policyholder behavior over the years. The proportion of claims among new policies also decreased to 18.2%, reflecting a shift in either the nature of the new policies or a more cautious approach in their issuance. In contrast, the termination policies consistently filed claims at very low rates, peaking at 4.4% in 2016 but remaining low overall, which may indicate that most terminating policies were risk-free by their end.

Correlations

As we have 30 variables, a correlation matrix may be difficult to look at and to interpret. We need first to determine which variables are the most interesting to study and to keep.

For example `Date_last_renewal` and `Date_next_renewal` are highly correlated as they are set apart by exactly one year, so for the rest of the study we will only consider `Date_last_renewal`.

The Length and Weight of a vehicle are also correlated enough (0.83) so we won't use the Length in the matrix, especially since it has many NA values.

We start first by displaying a matrix of correlations between variables related to the car, the policy holder and some policy attributes (`Premium`, `N_claims_year` and `Cost_claims_year`). The most correlated variables will be ordered from top to bottom, right to left (highest correlations in the top-right) :

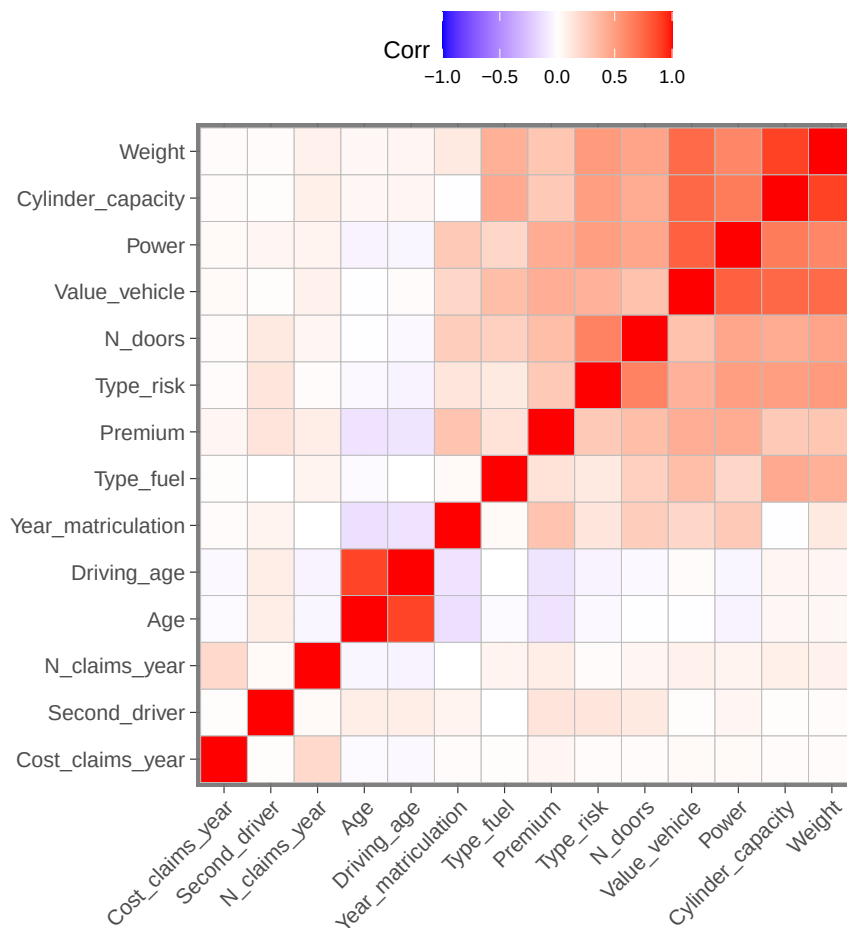


Figure 1: Correlation Matrix Car Details

The correlation matrix indicates that the car-related variables (`Power`, `Cylinder_capacity`, `Weight`, `Value_vehicle`, `N_doors`, and `Type_risk`) exhibit strong correlations with one another. Additionally, the `Premium` variable shows significant correlation with these variables. A notable correlation also exists between `Driving_age` and `Age`, which can be explained by the fact that most people take their driving licence at the same age, around 20.

Surprisingly the variables associated with claims (`N_claims_year`, and `Cost_claims_year`) show lower correlations with `Premium` or vehicle related characteristics.

The second matrix will show correlations between policy variables and Power, Value_vehicle and Type_risk. Once again, the highest correlations are at the top-right :

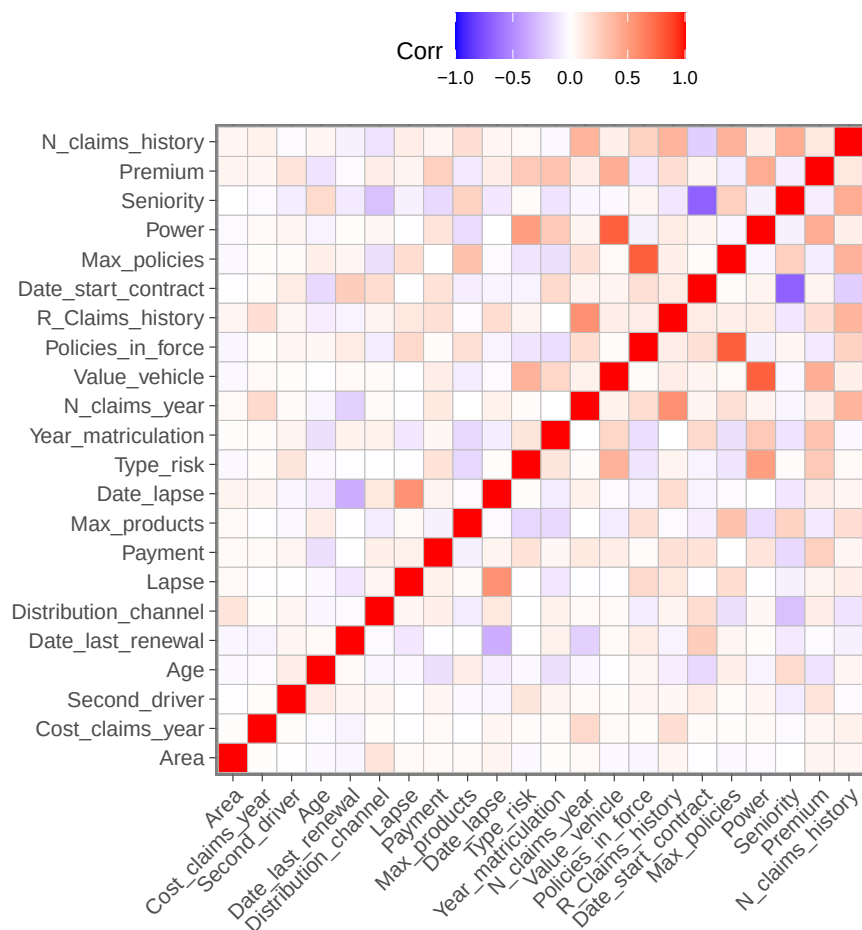
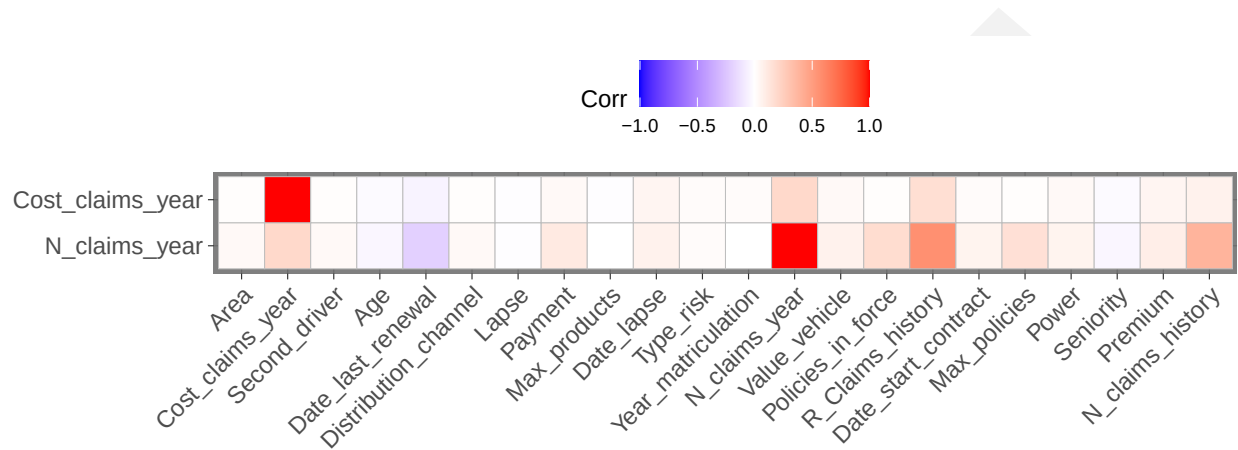


Figure 2: Correlation Matrix Policy Details

Strong positive correlations include the relationship between Power and Value_vehicle (0.777), suggesting that as a vehicle's power increases, its value tends to rise correspondingly. Similarly, R_Claims_history and N_claims_year exhibit a significant positive correlation (0.566), indicating that individuals with a more extensive claims history are likely to have a higher number of claims annually.

Conversely, the matrix reveals strong negative correlations, notably between Seniority and Date_start_contract (-0.678). This suggests that individuals with longer seniority tend to start their contracts earlier. Additionally, the correlation between Lapse and Date_lapse (-0.557) highlights that a higher lapse rate correlates with a longer duration since the last lapse occurred.

The figure below shows the correlation between N_claims_year (number of claims in a year), Cost_claims_year (total cost in a year) with the other variables. It is an extract of the previous matrix :



As we can see, the N_claims_year variable shows a strong positive correlation of 0.5659 with R_Claims_history, indicating that a higher number of claims correlates with an increased historical claim rate. Additionally, N_claims_year exhibits a significant correlation of 0.3925 with N_claims_history, suggesting that those with a higher number of past claims are likely to continue having claims in the current year.

In contrast, Cost_claims_year is positively correlated with N_claims_year at 0.1990, implying that as the number of claims in the current year increases, so does the cost attributed to those claims. Interestingly, the correlations with some variables are quite low, such as Premium, which shows a correlation of 0.054 with Cost_claims_year, indicating a minimal relationship.

As per correlation matrix suggests, the risk of claims and associated cost is mainly related to the vehicle itself. We will now see how the Power and Value of a vehicle can affect the risk of raising a claim :

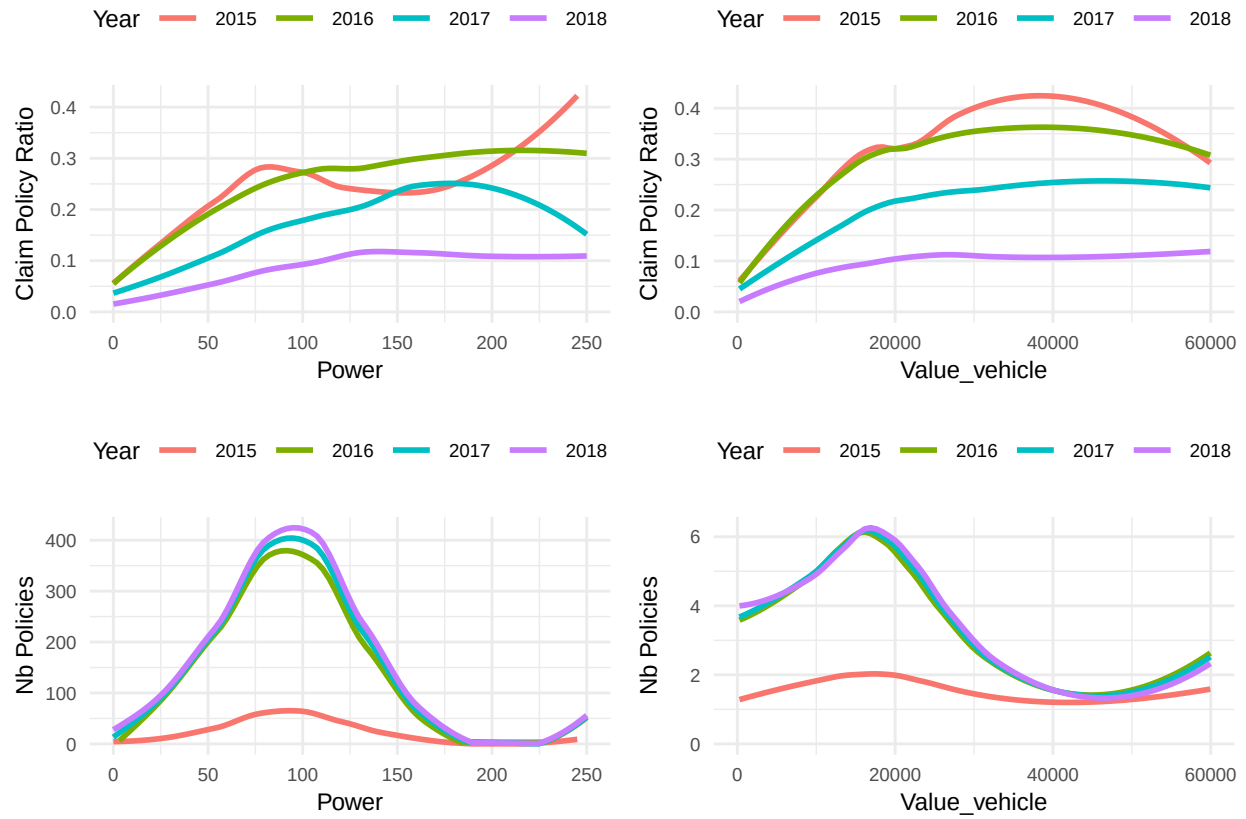


Figure 3: Claim Policy Ratio compared to Power, Value vehicle, Age and Driving age

The figure below shows the risk for a driver to raise a new claim based on the number of claims already raised. It doesn't factor the claim history, but only the claims raised in a year. We also don't display the risk for number of claims over 20 as the numbers are quite low.

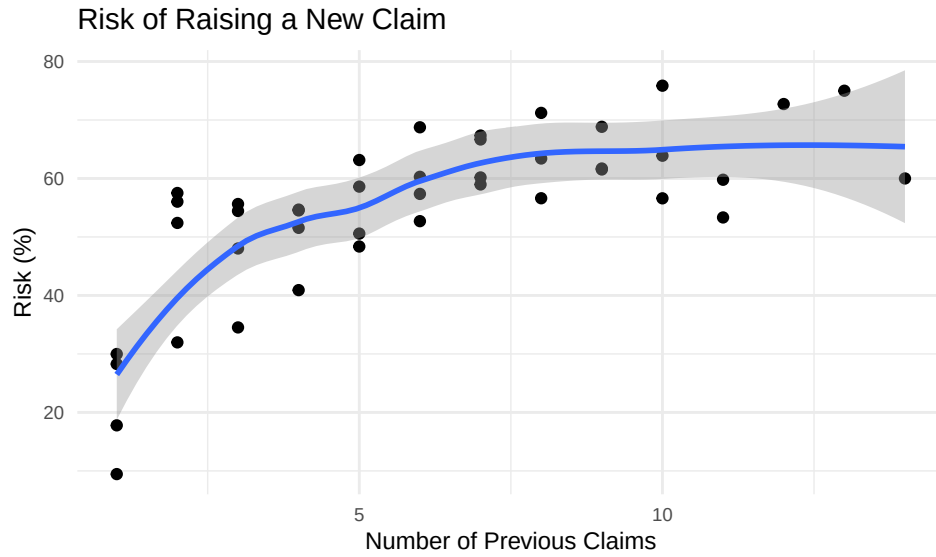


Figure 4: Risk of Raising a New Claim

The logic is as follow : each driver who raises N claims in a year have necessary raised (N-1) claims in that year before raising the Nth one. If we compare with the number of drivers who raised exactly N-1 claims, we can calculate the proportion of drivers who will raise N claims compared to the total drivers who raised N-1 claims :

$$Risk = \frac{Claims_n}{Claims_n + Claims_{n-1}} \quad (1)$$

If we now plot the number of claims raised in 2017 with the number of previous claims, we see that while there is a slow progression we have 2 trends : people who previously raised some claims and don't raise many new ones (below the blue line), and people who consistently raise new claims (above the blue line).

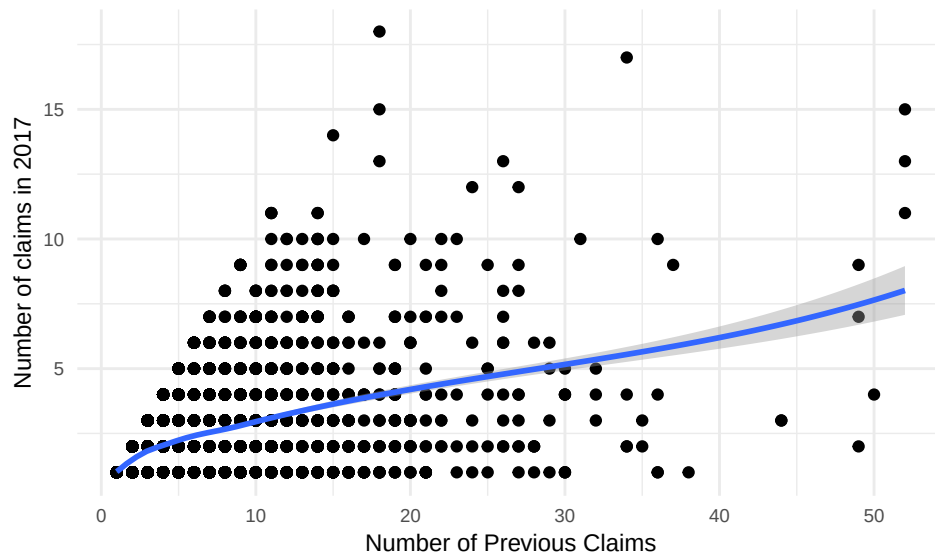


Figure 5: Risk of Raising a New Claim (2017)

These two previous graphs may indicate that some people will learn from their mistake and raise less claims when they have a history, and reckless drivers who consistently raise claims.

We now need to distinguish the 2 groups.

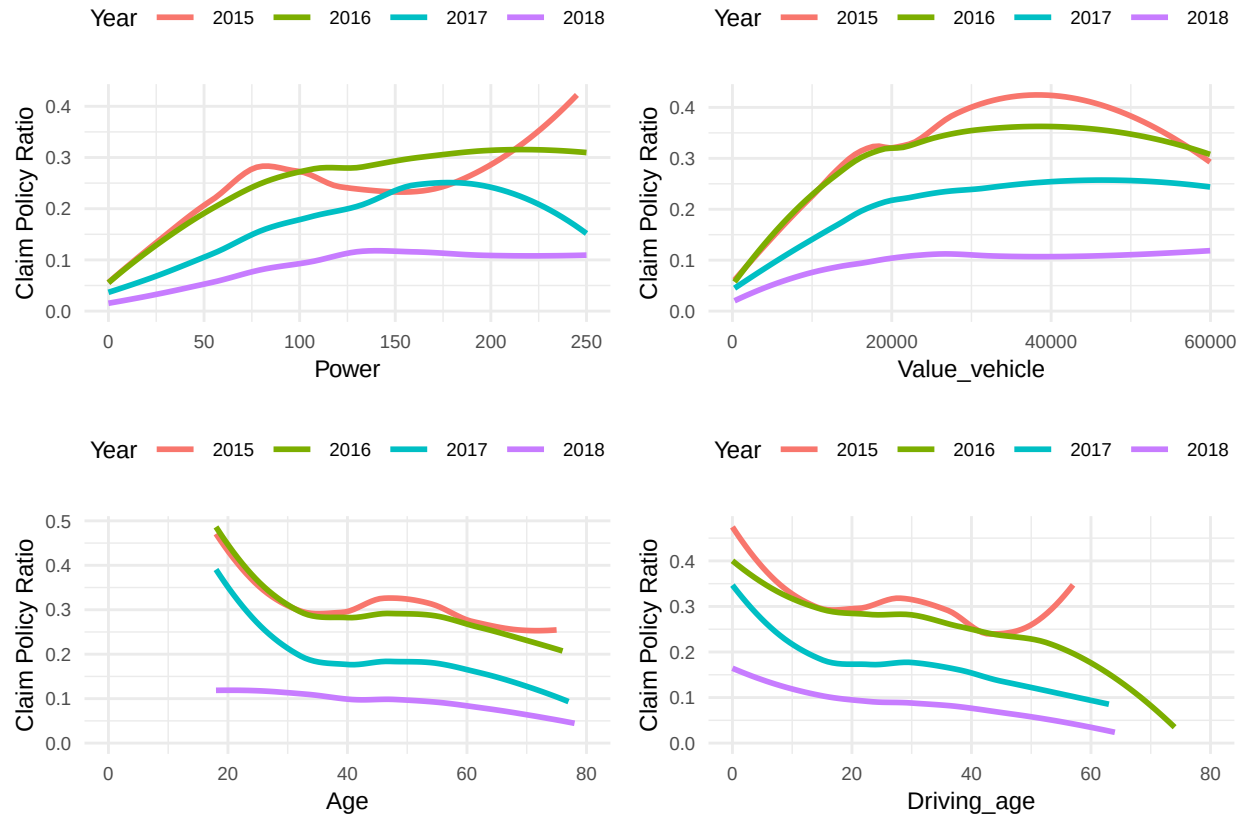


Figure 6: Claim Policy Ratio compared to Power, Value vehicle, Age and Driving age

The provided plots illustrate the relationship between various factors—power, vehicle value, age, and damage age—and their respective claim ratios over three distinct years.

Upper Left Plot: Power vs. Claim Ratio

The upper left plot shows that power has a significant effect on the claim ratio. As power levels increase, the claim ratio initially rises, peaking at moderate power levels before declining sharply. This suggests that both low-powered and very high-powered vehicles may be less associated with claims, while those in the mid-range may reflect a higher tendency to file claims. The trend remains consistent across the years, highlighting a stable relationship between vehicle power and insurance risk.

Upper Right Plot: Vehicle Value vs. Claim Ratio

In the upper right plot, vehicle value is analyzed against claim ratios. Here, the trend indicates that as vehicle value increases, the claim ratio tends to decline. This negative correlation suggests that more expensive vehicles may be better insured or that drivers of higher-value vehicles are perhaps more cautious, reducing the likelihood of incidents. The year-on-year comparison reveals a consistent downward trend, indicating that this relationship remains stable over time.

Lower Left Plot: Age vs. Claim Ratio

The lower left plot examines the impact of the vehicle's age on claim ratios. Interestingly, there is a clear pattern where newer vehicles generally have higher claim ratios, which may be attributed to the fact that recent models might be driven more often and thus expose them to a greater risk of incidents. As vehicles age, the claim ratio decreases, suggesting that older vehicles might be driven less frequently or perhaps are less involved in accidents, reflecting more cautious driving behavior.

Lower Right Plot: Damage Age vs. Claim Ratio

Lastly, the lower right plot focuses on damage age and its relationship with claim ratios. The graph reveals that as damage age increases, the claim ratio tends to decline. This might imply that older claims are less likely to result in payout when the vehicle has been in service for a longer duration since the initial damage. The trend is consistent across all years, reinforcing that the age of damage has a predictive quality regarding how likely a claim will be submitted.

Overall, these plots collectively enhance our understanding of how various characteristics of vehicles influence the likelihood of claims, providing valuable insights for insurance companies in risk assessment and policy pricing strategies.

DRAFT

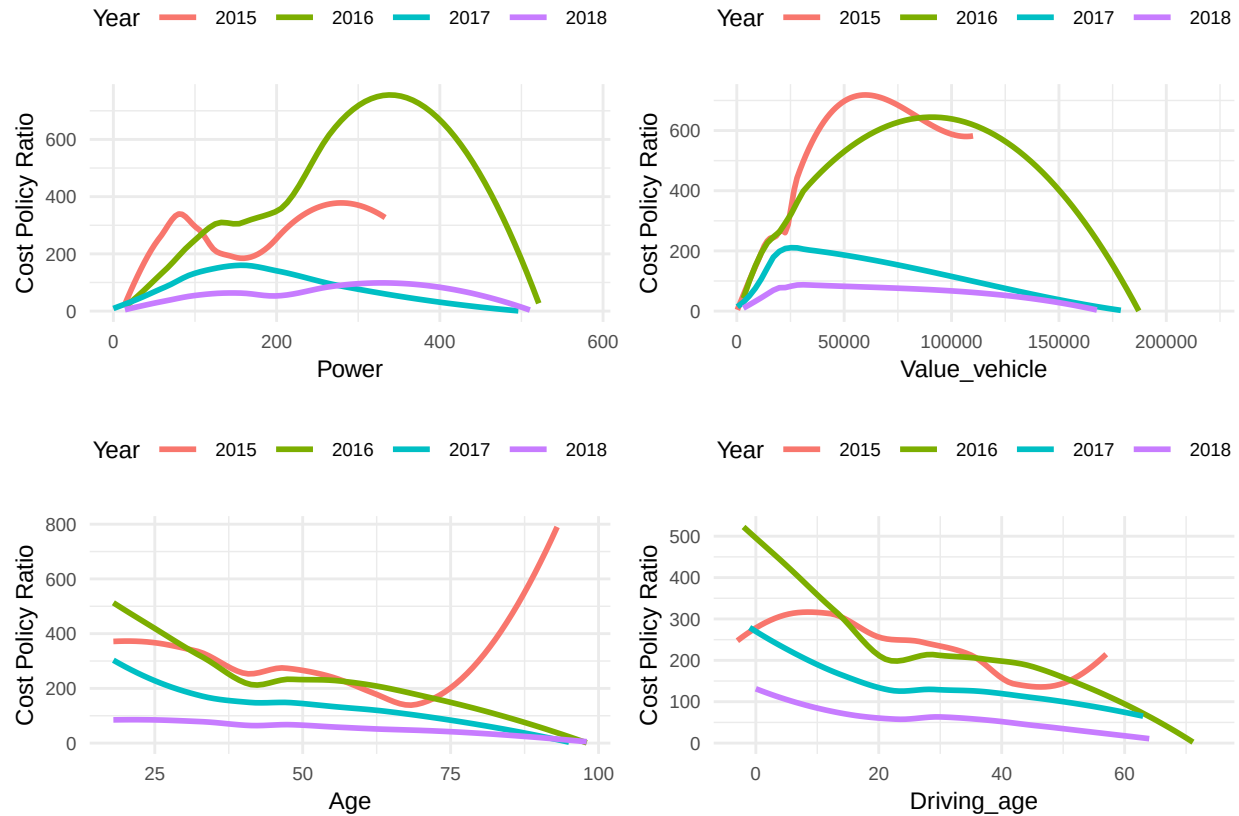


Figure 7: Cost Policy Ratio compared to Power, Value vehicle, Age and Driving age

The provided plots illustrate the relationship between several vehicle characteristics—Power, Value per Vehicle, Age, and Driving Age—and the Cost per Ratio over multiple years (from 2015 to 2018). Each plot offers insights into how these factors have influenced costs within the given timeframe. Top Left Plot: Power vs. Cost per Ratio

In the top left plot, the Power of vehicles shows a dynamic relationship with the Cost per Ratio over the years. As Power increases, the cost initially rises sharply but then exhibits a noticeable decline after reaching a peak. This suggests that while higher-power vehicles may have resulted in increased costs initially, there seems to be a market adjustment leading to lower costs for more powerful vehicles in subsequent years. The lines for different years vary significantly, indicating that the trend in cost related to power has changed over time, possibly due to advancements in technology or changes in consumer preferences. Top Right Plot: Value per Vehicle vs. Cost per Ratio

The top right plot focuses on the Value per Vehicle and its correlation with Cost per Ratio. Here, an upward trend is observed across the years, indicating that as the value of vehicles increases, so too does the cost per ratio. This may point to a market trend where more valuable vehicles are associated with higher costs, possibly due to enhanced features or technology. Notably, the line for 2018 remains consistently above those of earlier years, suggesting a continued escalation in costs related to higher-value vehicles. Bottom Left Plot: Age vs. Cost per Ratio

In the bottom left plot, Age is examined against the Cost per Ratio. The plot reveals a negative correlation; as the age of vehicles increases, the cost per ratio decreases. This could be attributed to depreciation, where older vehicles naturally carry less value, leading to lower costs. The lines convey a consistent downward trend across all years, emphasizing that age remains a stable predictor of cost efficiency in the automotive market. Bottom Right Plot: Driving Age vs. Cost per Ratio

The bottom right plot depicts the relationship between Driving Age and Cost per Ratio. Interestingly, the driving age does not exhibit a significant correlation with cost. The lines converge much more closely than in the other plots, suggesting that changes in driving age have less impact on vehicle costs over the years. This may reflect societal norms around driving, insurance policies, and the relative independence of the driving age from other vehicle characteristics.

Year	Area	Nb_policies	Nb_policy_claims	Total_cost
2015	Rural	3,177	911	812,282.2
2015	Urban	1,382	456	380,682.7
2016	Rural	22,136	6,081	5,084,952.8
2016	Urban	9,292	2,807	2,664,611.7
2017	Rural	24,561	4,294	3,600,673.0
2017	Urban	9,192	1,710	1,347,302.6
2018	Rural	26,770	2,455	1,714,713.0
2018	Urban	9,045	932	603,523.4

The provided data summarizes insurance claims across rural and urban areas from 2015 to 2018. Each entry represents a specific year and area type, detailing the Nb_policy_claims (number of policies with at least one claim), Total_cost (total claims cost), Nb_policies (total number of policies), Claim_Policy_Ratio (ratio of claims to policies), and Cost_Policy_Ratio (ratio of total costs to policies). For instance, in 2015, rural areas had 911 policies with claims, leading to a total cost of 812,282 from a total of 3,177 policies, resulting in a Claim_Policy_Ratio of 0.287 and a Cost_Policy_Ratio of 256. Urban areas in the same year reported 456 claim policies out of 1,382, reflecting slightly higher ratios. Over the years, it is evident that the number of claims in rural areas fluctuated significantly, with notable spikes and declines, suggesting varying insurance activity or risk levels. Overall, the data highlights differences in claim behavior and expenditure trends between rural and urban settings.

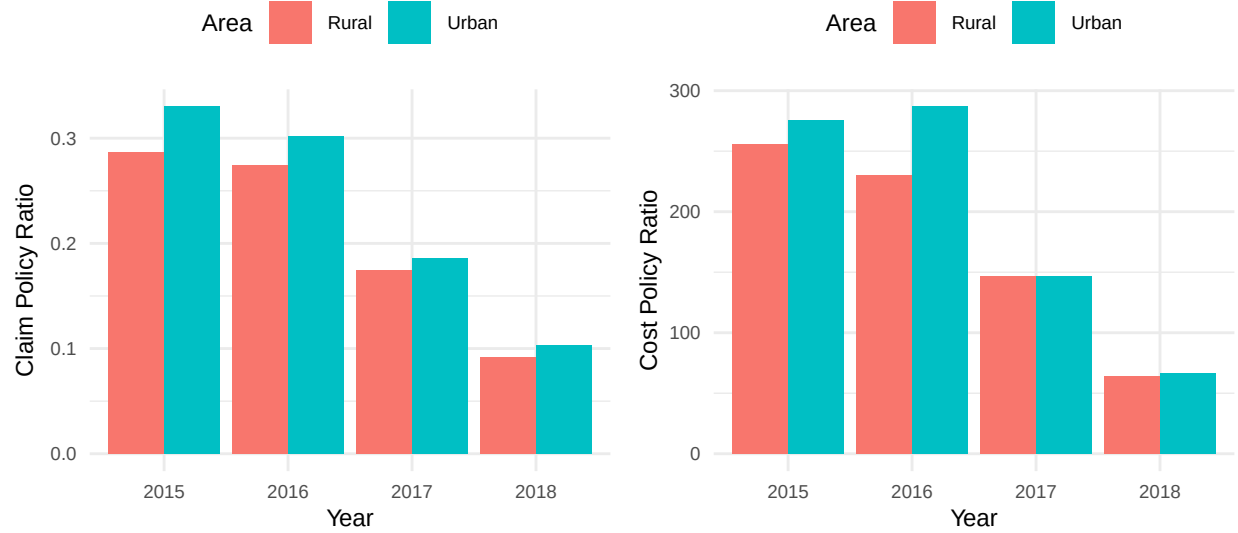


Figure 8: Claim Policy Ratio compared to Area : Nb of Claims and cost

Year	Type_risk	Nb_policies	Nb_policy_claims	Total_cost
2015	Motorbikes	218	25	14,105.84
2015	Vans	631	213	186,625.42
2015	Passenger cars	3,671	1,129	992,233.61
2015	Agricultural vehicles	39	0	0.00
2016	Motorbikes	2,424	269	106,047.28
2016	Vans	3,974	1,361	1,012,827.66
2016	Passenger cars	24,769	7,258	6,630,689.61
2016	Agricultural vehicles	261	0	0.00
2017	Motorbikes	2,777	220	143,896.22
2017	Vans	4,180	911	659,462.75
2017	Passenger cars	26,513	4,872	4,144,423.94
2017	Agricultural vehicles	283	1	192.67
2018	Motorbikes	3,083	130	45,006.90
2018	Vans	4,427	528	338,837.49
2018	Passenger cars	28,037	2,729	1,934,391.98
2018	Agricultural vehicles	268	0	0.00

The data reveals intriguing trends in car insurance across various vehicle types over several years. For motorbikes, there is a noticeable fluctuation in claims, with a generally low ratio of claims to policies, suggesting that while the risk remains, it hasn't translated into a high number of claims proportionate to the coverage offered. Conversely, vans show a higher ratio, particularly in certain years, indicating that they may pose a greater risk or that drivers are more likely to file claims.

Passenger cars consistently generate the highest number of claims, reflecting their prevalence on the road and possibly the higher likelihood of incidents associated with them. Agricultural vehicles, on the other hand, appear to be a much lower risk, with little to no claims reported across the years. This data highlights the complexities of risk assessment in insurance, as different vehicle types display distinct patterns of claims activity, influencing premium rates and underwriting strategies. Understanding these dynamics is essential for effectively managing risk in the insurance industry.

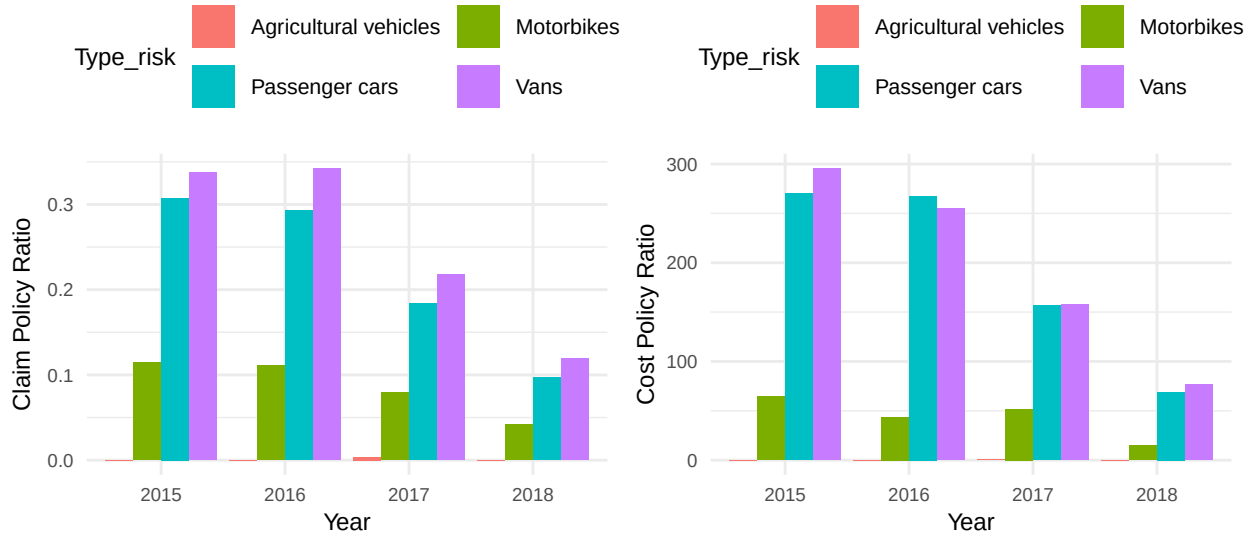


Figure 9: Claim Policy Ratio compared to Type of Vehicle : Nb of Claims and cost

References

- [1] Thomas Poufina, Periklis Gogas, Theophilos Papadimitriou and Emmanouil Zaganidis. 2023. “Machine Learning in Forecasting Motor Insurance Claims.” (Machine Learning and Artificial Intelligence in Non-life Insurance: Theory, Methods and Applications.) <https://www.mdpi.com/2227-9091/11/9/164>
- [2] Rafael Irizarry. “Introduction to Data Science.” <https://rafalab.dfci.harvard.edu/dsbook/>
- [3] Josep Lledó, Jose M. Pavía. 2024. “Dataset of an Actual Motor Vehicle Insurance Portfolio.” <https://data.mendeley.com/datasets/5cxyb5fp4f/2>
- [4] Dataset copied to GitHub. Available at: <https://github.com/jmr-lab/PH125.9x-Data-Science-CarInsurance> (Accessed: 2026).
- Glen, Stephanie. 2020. “RMSE: Root Mean Square Error.” StatisticsHowTo.com. <https://www.statisticshowto.com/probability-and-statistics/regression-analysis/rmse-root-mean-square-error/>
- Renata Balasco. 2026. “Factors Affecting Car Insurance Rates.” <https://www.thezebra.com/auto-insurance/driver/other-factors/>